Action-kernel behavioural model

fit_action_kernel / IBL `behavior_models.ActionKernel`

prior-mech-analysis notes

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1 Context

For act-conditioned analyses we currently estimate priors with a **fixed** learning rate $\alpha = 0.2$ via `action_kernel_priors`, applied the same way on every session before pooling into the supersession.

`fit_action_kernel` (in `scripts/simulate_synthetic_choices.py`) instead fits the full IBL `ActionKernel` choice model by MCMC on one session's real behaviour and returns a posterior-mean parameter vector. Open question (journal 2026-07-20): whether those session-specific fits should replace the fixed-α prior for act labels before supersession pooling.

Data. Fitting uses

$$(\text{actions},\ \text{stimuli},\ \text{stim\_side}) = \texttt{format\_data}(\texttt{trials}),$$

i.e. choices $a_t \in \{-1, +1\}$ ($0 =$ no-go), signed contrast $c_t \in [-1, 1]$, and stimulus side $s_t \in \{-1, +1\}$. Task `probabilityLeft` is *not* a covariate in this likelihood: the prior is the evolving action kernel.

2 Parameters (`single_zeta=True`)

Symbol	Code name	Role	Bounds
α	`alpha`	Learning rate of the action-history prior	$[0, 1]$
ζ	`zeta`	Sensory noise (shared across sides)	$[0, 1]$
ℓ_+	`lapse_pos`	Lapse when stim side is left ($s_t > 0$)	$[0, 0.5]$
ℓ_-	`lapse_neg`	Lapse when stim side is right ($s_t < 0$)	$[0, 0.5]$

Posterior mean $\hat{\boldsymbol{\theta}} = (\hat{\alpha}, \hat{\zeta}, \hat{\ell}_+, \hat{\ell}_-)$ is what `get_parameters(parameter_type="posterior_mean")` returns after MCMC.

3 Model equations

Notation (probabilities). Three different probabilities appear below—do not conflate them:

- π_t^{L} (and p_t in the analysis helper): **prior belief** that the next / current trial favours left, from the action-kernel EMA over past choices. *Not* the instantaneous choice probability.
- $\tilde{P}(\mathrm{L}_t)$: **pre-lapse probability of choosing left** given current contrast c_t and prior π_t^{L}.
- $P(\mathrm{L}_t)$: **probability of choosing left after lapses** (what is scored in the likelihood and sampled in simulation). Then $P(\mathrm{R}_t) = 1 - P(\mathrm{L}_t)$.

3.1 Action-kernel prior

Let $\boldsymbol{\pi}_t = (\pi_t^{\mathrm{R}}, \pi_t^{\mathrm{L}})$ be a probability vector over {right, left}, initialised at $\boldsymbol{\pi}_0 = (1/2, 1/2)$. With learning rate α,

$$\boldsymbol{\pi}_t = \begin{cases} (1-\alpha)\,\boldsymbol{\pi}_{t-1} + \alpha\,\mathbf{e}_{a_{t-1}} & \text{if } a_{t-1} \neq 0, \\ \boldsymbol{\pi}_{t-1} & \text{if } a_{t-1} = 0 \text{ (no-go)}, \end{cases} \tag{1}$$

where $\mathbf{e}_a$ is the one-hot indicator of the previous choice ($a = +1 \Rightarrow$ left, $a = -1 \Rightarrow$ right). The scalar left-prior used below is $\pi_t^{\mathrm{L}} = \boldsymbol{\pi}_t[1]$.

Range. The continuous left-prior is a probability:

$$\pi_t^{\mathrm{L}} \in (0,1) \quad \text{(equivalently } p_t \in (0,1) \text{ in (2))}.$$

It is initialised at $1/2$ and stays in $(0,1)$ under the EMA update (convex combination of a probability and $\{0,1\}$). In `ActionKernel`, values are additionally clipped to $[10^{-7}, 1-10^{-7}]$ before entering (3) (so $\Phi^{-1}(1-\pi)$ stays finite). For act L/R *splits* in analysis, `action_kernel_priors` binarises the continuous prior to the discrete labels $\{0.8, 0.2\}$ ($\geq 1/2 \mapsto 0.8$, else 0.2)—those are conditioning labels, not the continuous π_t^{L} used inside the fitted choice model.

Analysis helper (fixed α only). Our `action_kernel_priors` implements the same EMA on a scalar p_t:

$$p_0 = \tfrac{1}{2}, \qquad p_t = \alpha\,\mathbf{1}[a_{t-1} > 0] + (1-\alpha)\,p_{t-1}, \tag{2}$$

then binarises $p_t \mapsto \{0.8, 0.2\}$ for L/R-prior splits. It does *not* fit ζ or lapses.

3.2 Sensory likelihood combined with prior

Signed contrast c_t is combined with the prior belief π_t^{L} under sensory noise ζ (`combine_lkd_prior`; fixed $\sigma = 0.49$) to produce the *choice* probability before lapses:

$$\begin{aligned} \sigma^\star &= \left(\sigma^{-2} + \zeta^{-2}\right)^{-1/2}, \\ \tilde{P}(\mathrm{L}_t) &\equiv \tilde{P}(\text{choose left} \mid c_t, \pi_t^{\mathrm{L}}, \zeta) \\ &= \Phi\left(\frac{c_t}{\zeta} - \frac{\zeta}{\sigma^\star}\,\Phi^{-1}\big(1-\pi_t^{\mathrm{L}}\big)\right), \end{aligned} \tag{3}$$

where Φ is the standard normal CDF. (Equivalently $\tilde{P}(\mathrm{L}_t) = \Phi\big(c_t/\zeta + (\zeta/\sigma^\star)\Phi^{-1}(\pi_t^{\mathrm{L}})\big)$ since $\Phi^{-1}(1-\pi) = -\Phi^{-1}(\pi)$.)

3.3 Lapses

Side-dependent lapse rate

$$\ell_t = \begin{cases} \ell_+ & s_t > 0 \text{ (stim left)}, \\ \ell_- & s_t < 0 \text{ (stim right)}, \\ \tfrac{1}{2}(\ell_+ + \ell_-) & s_t = 0, \end{cases} \tag{4}$$

mixes the pre-lapse *choice* probability with chance:

$$P(\mathrm{L}_t) \equiv P(\text{choose left} \mid c_t, \pi_t, \boldsymbol{\theta}) = (1 - \ell_t)\, \tilde{P}(\mathrm{L}_t) + \frac{\ell_t}{2}. \tag{5}$$

With probability $1 - \ell_t$ the animal follows $\tilde{P}$; with probability ℓ_t it guesses 50/50 (hence the $\ell_t/2$ term). Always $P(\mathrm{R}_t) = 1 - P(\mathrm{L}_t)$.

3.4 Likelihood

For observed choices $\{a_t\}$,

$$\mathcal{L}(\boldsymbol{\theta}) = \prod_{t:\, a_t \in \{\pm 1\}} P(a_t \mid c_t, \boldsymbol{\pi}_t, \boldsymbol{\theta}). \tag{6}$$

MCMC targets the posterior of $\boldsymbol{\theta}$ under this likelihood (box priors on the bounds above). Early stopping uses Gelman–Rubin diagnostics across chains.

4 Simulation (`simulate_choices`)

Given fixed (c_t, s_t) and fitted $\hat{\boldsymbol{\theta}}$, `ActionKernel.simulate` runs the same forward pass (1)–(5), but updates $\boldsymbol{\pi}_t$ from **previously simulated** choices (generative), and samples $a_t \sim \mathrm{Bernoulli}\big(P(\mathrm{L}_t)\big)$ each trial. Stim/side (hence the empirical block schedule) stay fixed; only choices are random.

5 Relation to current analysis pipeline

- **Act prior labels (current):** (2) with $\alpha = 0.2$ on each session, then pool.
- **ActKernel choice null (`--actkernel-choice-null`):** fit $\boldsymbol{\theta}$ per `eid`, then `simulate_choices` on that session's real (c_t, s_t); re-split eligible trials under the same stim$\times$block conditioning.
- **Open:** replace fixed-α labels with session-fitted $\hat{\alpha}$ (or full kernel-derived priors) before supersession pooling.

References

- Package: `https://github.com/int-brain-lab/behavior_models` (`ActionKernel`, `utils.combine_lkd_pr`
- Local wrapper: `scripts/simulate_synthetic_choices.py` (`fit_action_kernel`, `simulate_choices`).
- Analysis prior helper: `block_analysis_allsplits.action_kernel_priors` ($\alpha = 0.2$).